MEDB 1.2 Crew Medical Officer Health Status Evaluations

3.2 Medical Requirements Overview

TABLE 3.2: MEDICAL REQUIREMENTS OVERVIEW

MEDB# and Title:	MEDB 1.2 Crew Medical Officer Health Status Evaluations
Sponsor:	Medical Operations
Discipline:	N/A
Category:	Medical Requirements
References:	SSP 50260 ISS Medical Operations Requirements Document (MORD) SSP 50667 Medical Evaluations Document (MED) Volume B
Purpose/Objectives:	To assess the medical status of the crewmember in-flight and report the findings to the Crew Surgeon.
Measurement Parameters:	Periodic Health Status (PHS) Evaluation: A. Vitals B. Physical Exam C. Labs- 1. Blood Chemistry (L+180 and per surgeon discretion) 2. Urinalysis (L+180 and per surgeon discretion)
Deliverables:	Data Collection Tool Form (and pictures if applicable)
Flight Duration:	≥ 30 days
Number of Flights:	Every ISS Increment
Number and Type of Crew Members Required:	All Crewmembers
Other Flight Characteristics:	N/A

3.3 Preflight Training

TABLE 3.3: PREFLIGHT TRAINING

Preflight Training Activity Description:	Two Crew Medical Officers (CMOs) per launch vehicle are trained to collect vital signs and body imagery in the launch minus 18/15 month time-frame.			
Schedule:	**Duration:**	**Schedule:**	**Flexibility:**	**Personnel Required:**
	CMO Skills 1 3 hours	L-18/15months	+/- 1 to 3 months	Instructor/ CMOs
Ground Support Requirements Hardware/Software:	**Preflight Hardware:**	**Preflight Software:**		**Test Location:**
	Station Support Computer (SSC) Medical Diagnostic Pack (Blood Pressure Device, Thermometer, Otoscope, Medical Supply Pack	Data Collection Tool (DCT) software		U.S
Training Facilities:	**Minimum Room Dimensions:**	**Number of Electrical Outlets:**	**Temperature Requirements:**	**Special Lighting:**
	Standard room, 8’ x 10’	4 (U.S. 110V, Russia 220V)	Normal, 20° – 25°C.	Normal lighting with dimming capability
	Hot or Cold Running Water:	**Privacy Requirements:**	**Other:**	
	Running water needed for hand washing.	Private room required	Internet connection for SSC	
Constraints/Special Requirements:	N/A			
Launch Delay Requirements:	Refresher training to be scheduled @ Crew Surgeon or crew request.			
Notes:	Training Procedures: SODF: ISS Medical Checklist			

3.4 Preflight Activities – No Preflight Activities

3.5 In-Flight Activities

TABLE 3.5.1: IN-FLIGHT ACTIVITIES

In-Flight Activity				
Description:	These tests will be conducted periodically in flight and as clinically indicated (ACI).			
	Duration:	**Schedule:**	**Flexibility:**	**Personnel Required:**
Schedule:	Setup: 10 minutes Exam: 15 minutes/CM CMO: 5 minutes/CM Stow: 10 minutes	6 month mission L+14/21 days L+90 days R-21/14 days ACI 1 year mission L+14/21 days L+90 days L+180 days L+270 days R-21/14 days ACI	+/- 3 days, with additional latitude to allow scheduling efficiency across multiple expedition crews.	CMO & Crewmembers
	Urinalysis (see MO-9)	L+180 days	+/- 3 days	CMO & 1 Year Crewmembers
	Blood Chemistry (see MO-10)	L+180 days	+/- 3 days	CMO & 1 Year Crewmembers
Procedures:	SODF: ISS Medical Checklist			
Constraints / Special Requirements:	-Schedule equipment setup prior to PHS. -Exercise or any strenuous activity that results in sore muscles should not be performed 24 hours prior to PHS. -Blood Chemistry and/or Urinalysis per Surgeon discretion for 6 month mission.			
Photo / TV Requirements:	Imagery is required (digital otoscope)			
Cold Stowage Requirements:	N/A			
Mission Extension Requirements:	N/A			

Data Delivery:	In-flight PHS discrete values are recorded in the DCT and will be saved along with PHS Otoscope images on the Station Support Computer (SSC) File Server. PHS data will be downlinked within the week of test completion from the SSC server. PHS data is archived within the Mission Extended Medical Enterprise (MEME) environment. Results for Urinalysis (MO-9) and Blood (MO-10) are forwarded securely via Kryptiq by the GMO to the Med Ops Data Distribution Specialist. The results are archived within the Mission Extended Medical Enterprise (MEME) Repository.

TABLE 3.5.2: IN-FLIGHT HARDWARE

Hardware/Software Name
Station Support Computer (SSC)
SSC Data Collection Tool Software
Medical Supply Pack
Medical Diagnostic Pack

3.6 Postflight Activities – No Postflight Activities

3.7 Summary Schedule

TABLE 3.7: SUMMARY SCHEDULE

ACTIVITY:	DURATION:	SCHEDULE:	FLEXIBILITY:	PERSONNEL REQUIRED:	CONSTRAINTS:
Preflight Training:					
CMO Skills 1	3 hours	L-18/15 months	+/- 1 to 3 months	Instructor/CMOs	N/A
Preflight: - N/A					
In-Flight:					
PHS	Setup: 10 minutes Exam: 15 minutes/CM CMO: 5 minutes/CM Stow: 10 minutes	6 month mission L+14/21 days L+90 days R-21/14 days ACI 1 year mission L+14/21 days L+90 days L+180 days L+270 days R-21/14 days ACI	+/- 3 days, with additional latitude to allow scheduling efficiency across multiple expedition crews	CMO & Crewmembers	-Schedule equipment setup prior to PHS. -Blood Chemistry and/or Urinalysis per Surgeon discretion for 6 month mission. -Strenuous activity that results in sore muscles should not be performed 24 hours prior to PHS.
Urinalysis	(see MO-9)	L+180 days	+/- 3 days	CMO & 1 Year Crewmembers	
Blood Chemistry	(see MO-10)	L+180 days	+/- 3 days	CMO & 1 Year Crewmembers	
Postflight: – N/A					
Postflight Debrief:					
PHS	Included as part of the overall Med Ops debrief				